

Table 1: Method comparison on CORE4D-G1 (all 50 paper sequences), mean over cases. **SPIDER-CEM** (ours) vs **SBTO** and **GMR**. Physical contact/penetration are frame fractions from the same MuJoCo contact detector; max penetration is the deepest hand-object interpenetration. $\uparrow/\downarrow$: higher/lower is better; best per row in bold. Note GMR’s low penetration@5mm is an artifact of barely contacting the object (contact@5mm $\approx$ 0.2%), not clean contact: its max penetration (23.5 mm) still exceeds ours (12.0 mm).

Metric	SPIDER-CEM	SBTO	GMR
Phys. contact@5mm (%) $\uparrow$	72.5	34.4	0.2
Phys. penetration@5mm (%) $\downarrow$	7.6	36.6	4.6
Max phys penetration (mm) $\downarrow$	12.0	25.4	23.5
Obj speed max (m/s) $\downarrow$	1.288	1.763	1.834
Foot slip max (m) $\downarrow$	0.866	1.200	0.971